

Extending Mendelian Genetics for Two or More Genes

Dominance relationships, multiple alleles, and pleiotropy all have to do with the effects of the alleles of a single gene. We now consider two situations in which two or more genes are involved in determining a particular phenotype. In the first case, called epistasis, one gene affects the phenotype of another because the two gene products interact; in the second case, called polygenic inheritance, multiple genes independently affect a single trait.

Epistasis

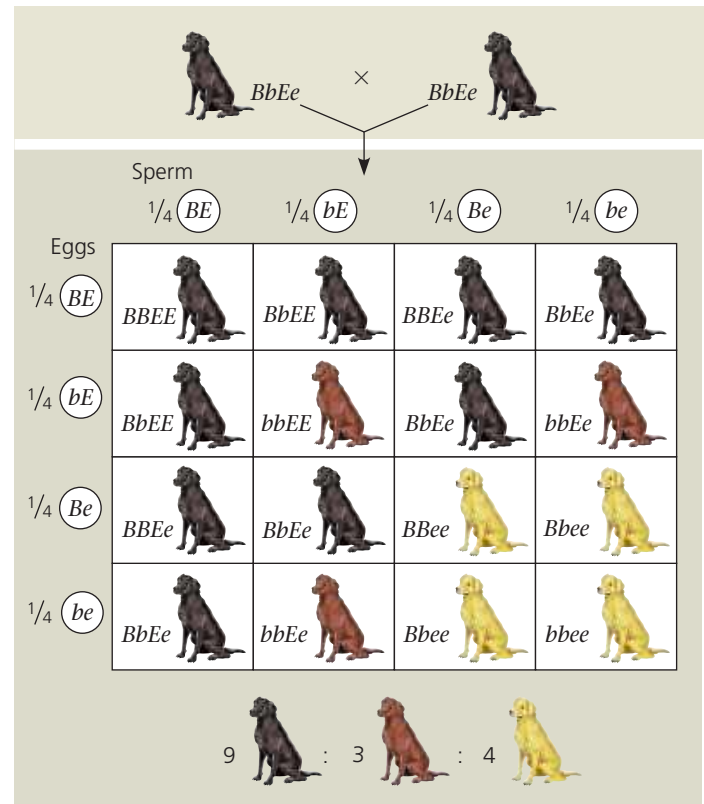
In **epistasis** (from the Greek for “standing upon”), the phenotypic expression of a gene at one locus alters that of a gene at a second locus. An example will help clarify this concept. In Labrador retrievers (commonly called “Labs”), black coat color is dominant to brown. Let’s designate B and b as the two alleles for this character. For a Lab to have brown fur, its genotype must be bb ; these dogs are called chocolate Labs. But there is more to the story. A second gene determines whether or not pigment will be deposited in the hair. The dominant allele, symbolized by E , results in the deposition of either black or brown pigment, depending on the genotype at the first locus. But if the Lab is homozygous recessive for the second locus (ee), then the coat is yellow, regardless of the genotype at the black/brown locus (yellow Labs). In this case, the gene for pigment deposition (E/e) is said to be *epistatic* to the gene that codes for black or brown pigment (B/b).

What happens if we mate black Labs that are heterozygous for both genes ($BbEe$)? Although the two genes affect the same phenotypic character (coat color), they follow the law of independent assortment. Thus, our breeding experiment represents an F_1 dihybrid cross, like those that produced a 9:3:3:1 ratio in Mendel’s experiments. We can use a Punnett square to represent the genotypes of the F_2 offspring (**Figure 14.12**). As a result of epistasis, the phenotypic ratio among the F_2 offspring is 9 black to 3 chocolate to 4 yellow Labs. Other types of epistatic interactions produce different ratios, but all are modified versions of 9:3:3:1.

Polygenic Inheritance

Mendel studied characters that could be classified on an either-or basis, such as purple versus white flower color. But many characters, such as human skin color and height, are not one of two discrete characters, but instead vary in the population in gradations along a continuum. These are called **quantitative characters**. Quantitative variation usually indicates **polygenic inheritance**, an additive effect of two or more genes on a single phenotypic character. (In a way, this is the converse of pleiotropy, where a single gene affects several phenotypic characters.) Height is a good example of polygenic inheritance: In 2014, a genomic study of over 250,000 people found almost 700 genetic variations in over 180 genes that affect height. Many variations were in or near

▼ **Figure 14.12 An example of epistasis.** This Punnett square illustrates the genotypes and phenotypes predicted for offspring of matings between two black Labrador retrievers of genotype $BbEe$. The E/e gene, which is epistatic to the B/b gene coding for hair pigment, controls whether or not pigment of any color will be deposited in the hair.



VISUAL SKILLS Compare the four squares in the lower right of this Punnett square with those in Figure 14.8. Explain the genetic basis for the difference between the ratio (9:3:4) of phenotypes seen in this cross and the 9:3:3:1 ratio seen in Figure 14.8.

genes involved in biochemical pathways affecting growth of the skeleton, but others were associated with genes not obviously related to growth. Another study in 2018, of 300,000 individuals, identified 124 genes that affect hair color. Although simplifying can sometimes be valuable in understanding genetic principles, as we’ll see shortly, the classic idea of a single gene determining eye color or hair color or other such traits is an oversimplification. (Earlobe attachment, an example used until a decade ago in this textbook, was shown in 2017 to be affected by nearly 50 genes!)

Skin pigmentation in humans is also controlled by many separately inherited genes—378 at latest count, many of which are involved in the production of melanin skin pigments. Here, we’ll simplify the story in order to understand the concept of polygenic inheritance. Let’s consider three genes, with a dark-skin allele for each gene (A , B , or C) contributing one “unit” of darkness (also a simplification) to the phenotype and being incompletely dominant to the other, light-skin allele (a , b , or c). In our model, an $AABBCC$ person would be very dark, whereas an $aabbcc$ individual would be very light.